Problem Statement ID	**26067**
Problem Statement Title	Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
Description	**Background** India's vast Exclusive Economic Zone (EEZ) and coastline demand continuous, high-resolution monitoring of ocean state variables. INCOIS routinely generates and archives large volumes of ocean model outputs - including three-dimensional fields of temperature, salinity, current vectors, chlorophyll, etc. - as well as real-time and delayed-mode observations from autonomous instruments such as Argo profiling floats and underwater Gliders. These datasets are stored in NetCDF and ASCII/text formats and span multiple depth levels, spatial grids, and time steps. Despite the richness of this data, no integrated, web-based 3D visualization platform currently exists that can simultaneously render model fields and in-situ instrument observations in a single interactive environment. Existing tools are either desktop-bound, support only 2D plan views, or lack the ability to co-visualize model outputs alongside instrument profiles. Operational oceanographers and forecasters are therefore forced to toggle between disparate software packages, making it difficult to rapidly correlate model predictions with observational evidence. **Key gaps identified include:** • No web-based, platform-independent 3D rendering of ocean model data (temperature, salinity, currents, etc.) with depth-resolved volumetric views. • No unified display of Argo float and Glider profile data (latitude, longitude, depth, time, temperature, salinity, chlorophyll) alongside model fields. • Absence of interactive controls for variable selection, depth-slice navigation, time-step animation, and customizable colorbars. • Inability to ingest new observational data streams or additional model variables without significant re-engineering. • Lack of tools to support intuitive, rapid understanding of complex 3D ocean phenomena for operational decision-making. The absence of such a system impedes timely hazard assessment, search-and-rescue support, fishery advisories, climate monitoring, etc. - all operational mandates of INCOIS. **Expected Solution** The proposed solution is a **web-based, browser-native 3D Ocean Data Visualization System** that integrates ocean model outputs with observational data on a single interactive platform. **Core functional requirements:** • **3D Volumetric Rendering:** Interactive visualization of ocean model fields (temperature, salinity, current vectors) across the full water column, with support for depth-slice views, isosurface extraction, and time-step animation using WebGL / Three.js or Cesium.js. • **Instrument Data Overlay:** Co-display of Argo float, Glider profile, CTD and BGC data using geospatially accurate markers; users can click a float/glider to inspect a depth-vs-variable profile chart with timestamps. • **Multi-format Data Ingestion:** Automated parsers for NetCDF (via PyNIO / xarray backend) and delimited text formats, with a modular architecture that allows new variables or data sources to be added with minimal code change.

	• **Customizable Colorbar & Variable Controls:** Dynamic colorbar editor (color palette, min/max range, log/linear scale), variable selector, layer opacity controls, and vertical exaggeration slider for intuitive depth perception. • **Web-based, Scalable Architecture:** Frontend built on modern JavaScript frameworks with a lightweight REST/OPeNDAP API backend, enabling deployment on INCOIS infrastructure without any client-side dependencies. • **Extensible Design:** Plugin-style module for future integration of additional sensors (e.g., CTDs, moorings, HF-radar, Acoustic Doppler Current Profiler (ADCP), etc.), new ocean model variables, and machine-learning derived products. The system will follow open standards (**OGC WMS/WCS, CF Conventions for NetCDF**), enabling interoperability with national and international ocean data portals. The end product will empower INCOIS forecasters to perform rapid, intuitive analysis of complex 3D ocean phenomena - significantly improving the speed and accuracy of operational advisories, in the same way that 3D meteorological visualization has transformed weather forecasting workflows. **Public Outreach & Science Communication** Beyond operational use, the platform will serve as a powerful science communication tool. Complex numerical ocean model outputs - which are typically inaccessible to non-specialists - can be transformed into visually intuitive, interactive 3D experiences. This makes the tool valuable for educating school and college students about ocean dynamics, engaging the general public during awareness campaigns, and supporting policymakers in understanding marine environmental conditions. INCOIS can use the platform for outreach events, exhibitions, and e-learning initiatives, bridging the gap between cutting-edge ocean science and the common person.
Organization	Ministry of Earth Sciences (MoES)
Department	Indian National Centre for Ocean Information Services (INCOIS) Ocean Valley
Category	Software
Theme	**Disaster Management**
Dataset Link	**The following dataset links are missing and should be included: a. Numerical Ocean Model Outputs: https://las.incois.gov.in/ & https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030/ description b. Argo Global Data: ftp://ftp.ifremer.fr/ifremer/argo c. Glider Data: ftp://ftp.ifremer.fr/ifremer/glider/v2/ d. Collection of In-situ Data:**